

Week 4 · Laboratory Problems — build the heading autopilot in Simulink

- Course: USV Guidance, Navigation and Control (Graduate) · Department of Autonomous Vehicle System Engineering
- Time: **one hour**, immediately after the Week 4 lecture hour
- Three problems, in order. Each one adds one branch to the previous answer.

What this hour is for

Week 3 closed a loop around a **speed** and found an error that no gain removed, no ringing, and a derivative term that only hurt. This hour closes the same loop around an **angle** and finds the opposite on all three counts. The difference lies in the axis, not in the controller.

The hull and the control allocation are provided. The loop that decides what yaw moment to demand is the exercise.

Before starting

```
cd lectures/W04_simulink/problems
W04_P1_start          % creates W04_P1.slx – hull and allocation only
W04_check(1)          % run this whenever, as often as needed
```

The solver is already set to fixed-step `ode4` at `h = 0.02` s. **Do not change it.**

★ Two requirements on every model

- A To Workspace block named `xlog`, format `Structure With Time`, fed by the plant's twelve-state output. The heading is state **12**.
- The blocks use the variable names `Kp`, `Ki`, `Kd`, `Nf`, `use_ssa`, `psi_step` and `t_step`. The checker sets them; a number typed into a block is not changed by the checker.

Problem 1 · Proportional only (20 minutes)

Build. A Step of `psi_step` degrees at `t_step`, converted to radians, the error, `ssa`, one gain, into the allocation's `tau_N`.

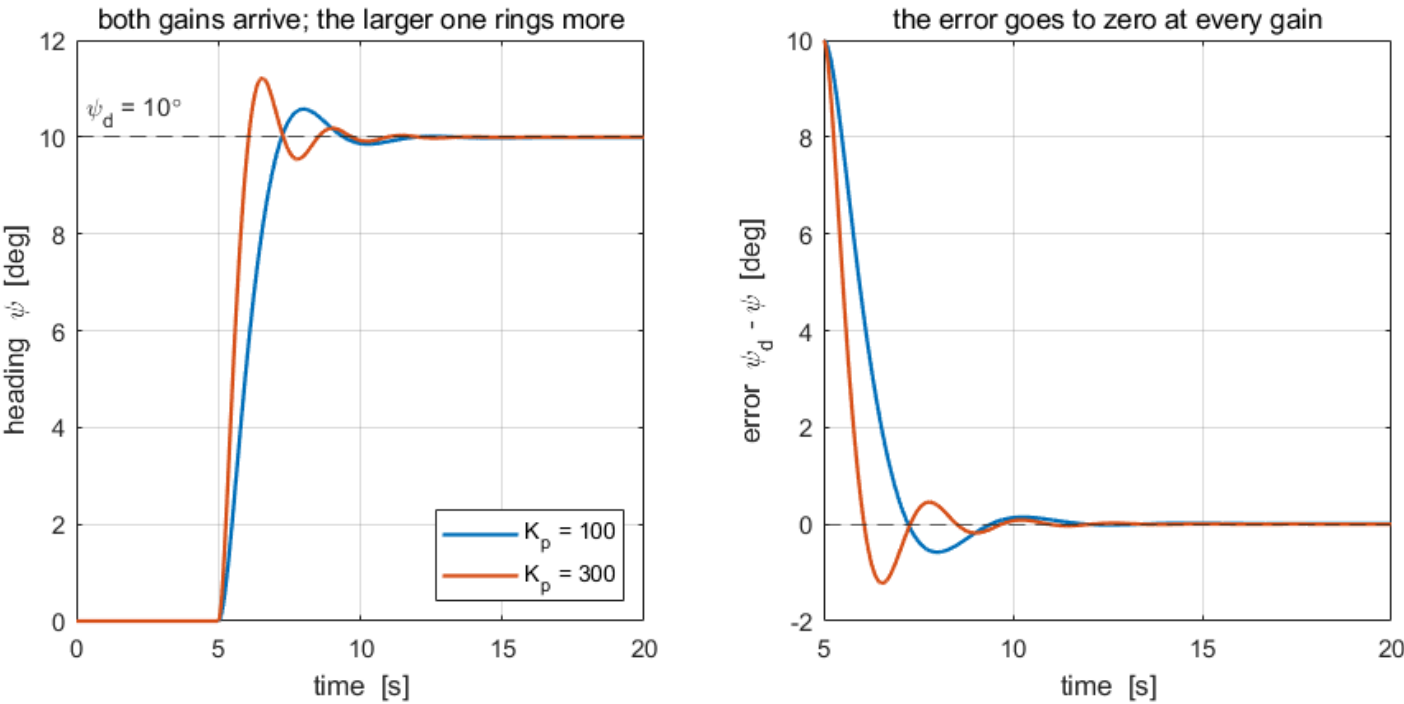
```
Step ψ_d [deg] → pi/180 → (+)(-) → ssa → Kp → τ_N → Control allocation → Otter USV
                    ↑                                     ↓
                    └────────────────── ψ (state 12) ─────────────────┘
```

Predict before running. In Week 3, P alone left a speed error at every gain. What error will P leave on the heading, and why?

Verify. `W04_check(1)`, a 10° step.

K_p	steady error	overshoot
100	0	5.8 %
300	0	12.2 %

What a correct model produces



Reading the figure	
left	both gains arrive at 10° ; the larger one arrives sooner and rings more
right	the same runs as error; both go to zero
the check	if a trace settles short of 10° , the feedback is not the heading — the Selector must pick state 12

The point. The heading is the running sum of the turn rate, so the plant already contains an integrator: the error goes to zero with P alone. Unlike the speed of Week 3, the heading also rings, because the heading, like the position of Week 2, can overshoot.

Problem 2 · The derivative (20 minutes)

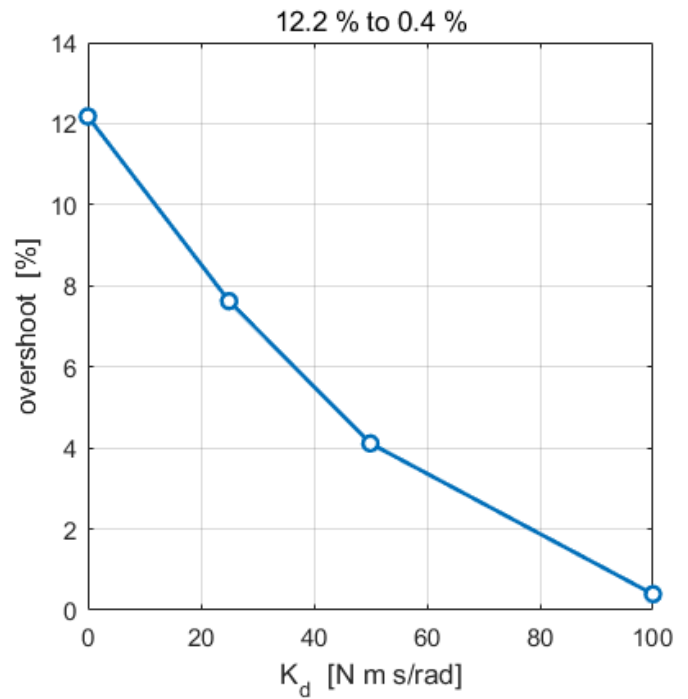
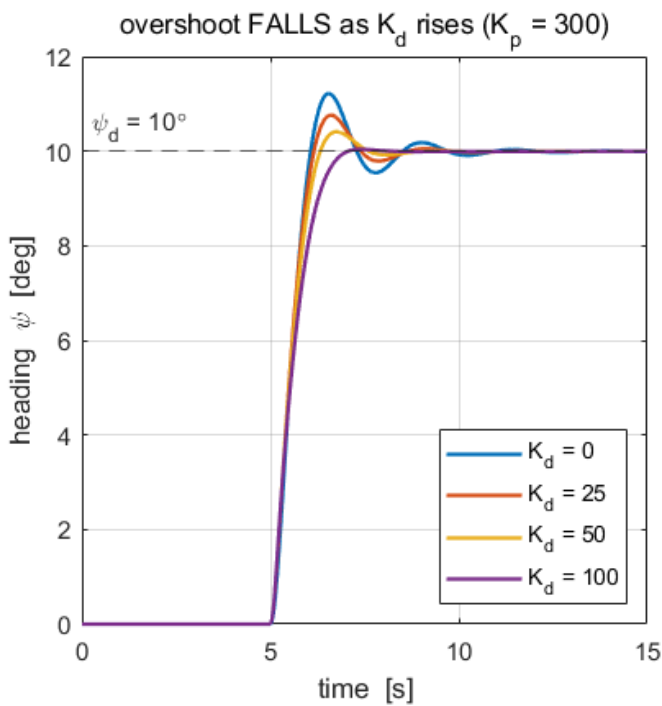
Build. One more branch: the error through **one Transfer Fcn**, numerator `[Kd*Nf 0]`, denominator `[1 Nf]`, added to the proportional term — the **D filter** of Week 2.

Predict before running. In Week 3 the derivative made the speed loop worse. The heading is an angle, like a position. Which way will the overshoot move as K_d rises?

Verify. `w04_check(2)`, $K_p = 300$, a 10° step.

K_d	overshoot
0	12.18 %
50	4.10 %
100	0.39 %

What a correct model produces



Reading the figure	
left	four responses to the same step; the overshoot falls as K_d rises
right	the overshoot against K_d
the check	if the overshoot rises with K_d the sign of the derivative branch is reversed

The point. On the heading the derivative is a damper again, as on the mass of Week 2. The term did not change between Weeks 3 and 4; the axis did.

Problem 3 · The wrap (20 minutes)

Build. A Switch driven by the Constant `use_ssa`: its upper input is the error through `ssa`, a Fcn block with `atan2(sin(u), cos(u))`; its lower input is the raw error.

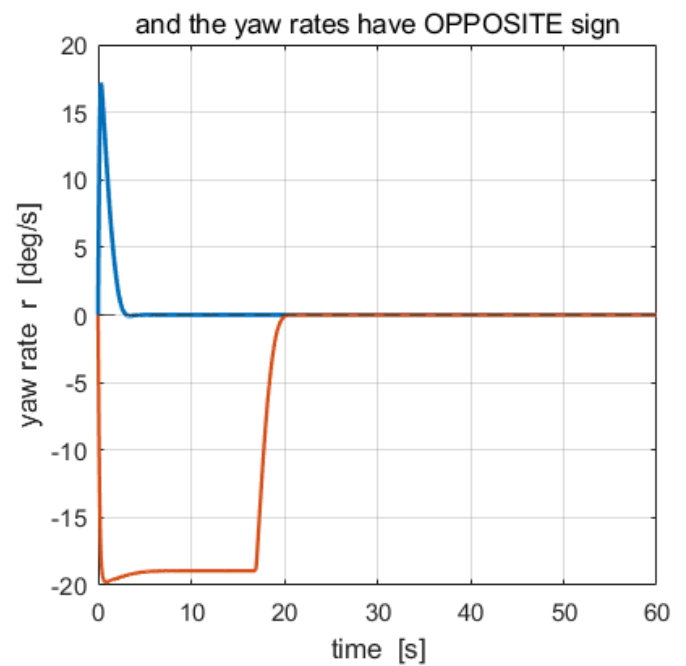
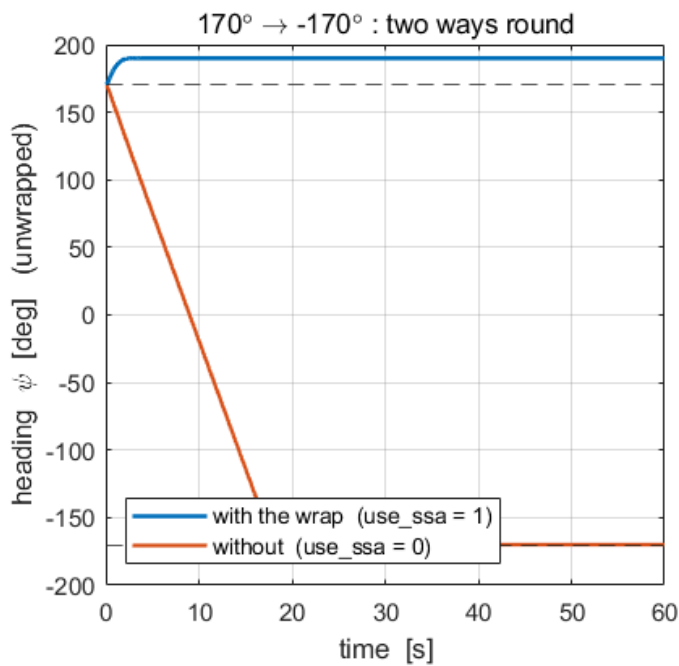
$$ssa(e) = \text{atan2}(\sin e, \cos e) \in (-\pi, \pi]$$

Set up the test. The checker starts the vessel at $\psi = 170^\circ$ and commands $\psi_d = -170^\circ$ from $t = 0$. The two headings are 20° apart.

Verify. `w04_check(3)`, $K_p = 300$, $K_d = 100$.

	turn executed
with the wrap	$+20^\circ$
without it	-340°

What a correct model produces



Reading the figure	
left	heading, unwrapped ; one run turns 20°, the other 340° the other way; both end at the same physical heading
right	the yaw rates have opposite sign for the whole manoeuvre
the check	if the two traces are identical, <code>use_ssa</code> is not reaching the Switch

The point. Without the wrap the error is computed as -340° and the vessel goes the long way round — seventeen times further, for the same commanded heading.

Marking

	Weight	What is being marked
Problem 1	30	<code>w04_check(1)</code> passes; why the error is zero is stated in one sentence
Problem 2	40	<code>w04_check(2)</code> passes; why the derivative damps here and hurt in Week 3 is stated
Problem 3	30	<code>w04_check(3)</code> passes; the cost of omitting <code>ssa</code> is quantified

If something goes wrong

Symptom	Cause	Fix
The model has no To Workspace block whose variable name is xlog	the variable name is still <code>simout</code>	rename it
the heading settles short of the command	the feedback is not state 12	the Selector index must be 12
the vessel spins continuously	the error sign is reversed	the sum is $\psi_d - \psi$
the overshoot rises with K_d	the derivative branch has the wrong sign	add it with $+$, the error already carries the sign
the two Problem 3 traces are identical	<code>use_ssa</code> never reaches the Switch	a Constant block with the value <code>use_ssa</code> , into the Switch's middle input
angles 57 times too large or small	degrees and radians mixed	every angle inside the loop is in radians; convert once, at the command

Reference answers are in `../solutions/`. Read them **after** attempting the problems.